

Matrix Theory Lecture Notes

Compiled for Exam Preparation

June 2026

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1 Lecture 01: Basics of Matrix and Determinant

Series: Matrix Theory **Lecture:** 01 of 09 **Video:** <https://www.youtube.com/watch?v=2NRTmf4SrII>

1.1 Prerequisites

- Basic algebra and arithmetic operations.
- Familiarity with basic mathematical notations.

1.2 1. Definition of a Matrix

A matrix is a rectangular array (arrangement) of numbers arranged in rows (horizontal) and columns (vertical). Matrices provide a compact way to represent and work with linear transformations and systems of equations. An $m \times n$ matrix contains m rows and n columns. The elements are denoted by a_{ij} , representing the entry in the i -th row and j -th column.

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}_{m \times n}$$

Examples: * **Example 1 (Square Matrix):** Number of rows equals number of columns (3×3).

$$A = \begin{bmatrix} 2 & 3 & 4 \\ 6 & -2 & 0 \\ 3 & 8 & -10 \end{bmatrix}$$

- **Example 2 (Rectangular Matrix):** Number of rows does not equal number of columns (2×3).

$$B = \begin{bmatrix} 2 & 3 & 4 \\ 5 & 6 & 7 \end{bmatrix}$$

1.3 2. Types of Matrices

- **Row Matrix:** A matrix with only one row ($1 \times n$).

$$A = [a_1 \quad a_2 \quad \dots \quad a_n]_{1 \times n}$$

- **Column Matrix:** A matrix with only one column ($n \times 1$).

$$B = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}_{n \times 1}$$

- **Zero (Null) Matrix:** A matrix where all elements are zero.

$$O = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

1.4 3. Matrix Determinants

The determinant is a scalar value calculated from a square matrix. While matrices are denoted using square brackets $[A]$ and represent arrays, determinants are denoted using straight vertical lines $|A|$ and represent a single numerical value.

1.4.1 Order 2 Expansion

For a 2×2 determinant, we cross-multiply:

$$\begin{vmatrix} 2 & 3 \\ 5 & -1 \end{vmatrix} = 2(-1) - 3(5) = -2 - 15 = -17$$

1.4.2 Order 3 Expansion

For a 3×3 determinant, we expand along a row or column using the sign convention grid (based on $(-1)^{i+j}$):

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

Example:

$$|A| = \begin{vmatrix} 5 & 6 & 1 \\ 0 & 2 & 7 \\ -1 & -2 & 6 \end{vmatrix}$$

Expanding along the first row (5, 6, 1):

$$\begin{aligned} |A| &= 5 \begin{vmatrix} 2 & 7 \\ -2 & 6 \end{vmatrix} - 6 \begin{vmatrix} 0 & 7 \\ -1 & 6 \end{vmatrix} + 1 \begin{vmatrix} 0 & 2 \\ -1 & -2 \end{vmatrix} \\ &= 5[12 - (-14)] - 6[0 - (-7)] + 1[0 - (-2)] \\ &= 5(26) - 6(7) + 1(2) \\ &= 130 - 42 + 2 = 90 \end{aligned}$$

(Note: Determinants are not directly added or subtracted element-wise like matrices; we generally evaluate their final numerical value first.)

1.5 4. Matrix Operations

1.5.1 Matrix Addition

Condition: The order of both matrices must be exactly equal. We add the corresponding elements.

Example:

$$\begin{bmatrix} 2 & 3 \\ 4 & 6 \end{bmatrix} + \begin{bmatrix} -1 & -2 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 2 + (-1) & 3 + (-2) \\ 4 + 0 & 6 + 3 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 4 & 9 \end{bmatrix}$$

1.5.2 Matrix Multiplication

Condition: Multiplication AB is only possible if the number of **columns in the first matrix** equals the number of **rows in the second matrix**. * If A has order $m \times n$ and B has order $n \times p$, multiplication AB is possible. * The resulting matrix AB will have order $m \times p$. * **Non-commutative:** Generally, $AB \neq BA$.

1.6 What Comes Next

2 Lecture 02: Rank of Matrix: A Concept

Series: Matrix Theory Lecture: 02 of 09 Video: <https://www.youtube.com/watch?v=h1FgjMdIEJ0>

2.1 Prerequisites

- Understanding of square matrices and determinants.
- Familiarity with expansion of determinants.

2.2 1. Minors and Rank of a Matrix

2.2.1 Minor of a Matrix

Let A be any matrix (square or rectangular). From this matrix A , delete all columns and rows leaving a certain p columns and p rows. Now if $p \geq 1$, then the elements which have been left constitute a square matrix of order p . The determinant of this square matrix is called a **minor** of A of order p .

For example, given a 3×3 matrix:

$$A = \begin{bmatrix} 3 & 2 & 5 \\ -1 & 0 & 6 \\ 7 & 8 & 3 \end{bmatrix}$$

The largest minor is of order 3:

$$|A| = \begin{vmatrix} 3 & 2 & 5 \\ -1 & 0 & 6 \\ 7 & 8 & 3 \end{vmatrix}$$

Examples of minors of order 2 include:

$$\begin{vmatrix} 3 & 2 \\ -1 & 0 \end{vmatrix}, \quad \begin{vmatrix} 2 & 5 \\ 0 & 6 \end{vmatrix}, \quad \begin{vmatrix} 3 & 5 \\ -1 & 6 \end{vmatrix}$$

(Note: Even for a 3×4 matrix, the largest possible minor is of order 3, because a minor must be a square determinant.)

2.2.2 Rank of a Matrix

Let A be any matrix. A number r is called the **rank** of the matrix A , denoted by $\rho(A)$, if it satisfies the following two properties: 1. There is at least one minor of A of order r which does not vanish (i.e., is not equal to zero). 2. Every minor of A of order higher than r vanishes (i.e., is equal to zero).

Important Properties of Rank: * The rank of a zero matrix is zero ($\rho(O) = 0$). * The rank of an identity/unit matrix I_n of order n is n . * The rank of a non-singular square matrix A of order n is n , because its determinant $|A| \neq 0$.

2.2.3 Nullity of a Matrix

Let A be a square matrix of order n . Then the number $(n - r)$, where r is the rank of A , is called the **nullity** of the matrix A .

$$N(A) = n - \rho(A)$$

2.3 2. Methods to Find the Rank of a Matrix

Equivalent matrices (matrices transformed by elementary operations) have equal ranks. There are three primary methods to find the rank: 1. **Determinant Method** (effective mostly for small matrices like 3×3). 2. **By Echelon Form of Matrix** (uses only **Row operations**). 3. **By Normal Form (Canonical Form) of Matrix**.

2.3.1 Echelon Form

A matrix is in Echelon form if: 1. All the non-zero rows, if any, precede the zero rows. 2. The number of zeroes preceding the first non-zero element in a row is less than the number of such zeroes in the succeeding row. 3. The first non-zero element in each row is unity (1).

Example of Echelon Form:

$$\begin{bmatrix} 1 & 5 & 6 & 7 & 0 & 8 \\ 0 & 1 & 2 & 5 & -1 & 2 \\ 0 & 0 & 0 & 1 & 2 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

[!NOTE] If a matrix is in Echelon form, the **rank of the matrix is equal to the number of non-zero rows** in it.

2.3.2 Normal Form (Canonical Form)

If A is a matrix of order $m \times n$ and rank r , then A can be reduced by the application of elementary transformations (both row and column) to any of the following block forms:

1. $\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$
2. $\begin{bmatrix} I_r \\ 0 \end{bmatrix}$
3. $\begin{bmatrix} I_r & 0 \end{bmatrix}$
4. $[I_r]$

where I_r is an identity matrix of order r , and 0 represents zero matrices of appropriate dimensions.

2.4 3. Numerical Problem 1: Rank by Echelon Form

Problem: Find the rank of the matrix:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{bmatrix}$$

Solution: We apply elementary row operations to reduce A to Echelon form.

Step 1: Use the 1 in the first row to zero out the elements below it. Apply $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - 2R_1$:

$$A \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & -1 \\ 0 & 2 & -1 \end{bmatrix}$$

Step 2: Use the second row to zero out the element below it. Apply $R_3 \rightarrow R_3 - R_2$:

$$A \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

The matrix is now in Echelon form (the leading element in row 2 can be made 1 by dividing, but it is not strictly necessary to determine the number of non-zero rows). There are exactly 2 **non-zero rows**.

Thus, the rank of the matrix is:

$$\rho(A) = 2$$

3 Lecture 03: Rank of Matrix By Determinant and Echelon Form

Series: Matrix Theory **Lecture:** 03 of 09 **Video:** <https://www.youtube.com/watch?v=DNFrFWb7RpM>

3.1 Prerequisites

- Understanding of matrix rank and minors.
- Familiarity with elementary row operations.

3.2 1. Problem 1: Determinant vs Echelon Form

Problem: Find the rank of the matrix using both the Determinant method and the Echelon form method:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{bmatrix}$$

3.2.1 Method 1: Determinant Method

Calculate the largest minor (the determinant of the 3×3 matrix):

$$|A| = \begin{vmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{vmatrix}$$

Expanding along the first row:

$$|A| = 1(20 - 12) - 2(5 - 4) + 3(6 - 8)$$

$$|A| = 8 - 2(1) + 3(-2) = 8 - 2 - 6 = 0$$

Since $|A| = 0$, the rank cannot be 3 ($\rho(A) < 3$). We now choose a minor of order 2:

$$M = \begin{vmatrix} 1 & 2 \\ 1 & 4 \end{vmatrix} = 4 - 2 = 2 \neq 0$$

Since a non-zero minor of order 2 exists, the rank is:

$$\rho(A) = 2$$

3.2.2 Method 2: Echelon Form Method

Apply elementary row operations to A : **Step 1:** $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - 2R_1$:

$$A \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & -1 \\ 0 & 2 & -1 \end{bmatrix}$$

Step 2: $R_3 \rightarrow R_3 - R_2$:

$$A \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

Step 3: Divide R_2 by 2 ($R_2 \rightarrow R_2/2$):

$$A \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -1/2 \\ 0 & 0 & 0 \end{bmatrix}$$

The matrix is now in Echelon form. * **Number of non-zero rows:** 2 * **Rank:** $\rho(A) = 2$ * **Nullity:** Order - Rank = $3 - 2 = 1$

3.3 2. Problem 2: Echelon Form of a 4×4 Matrix

Problem: Find the rank and nullity of:

$$A = \begin{bmatrix} 1^2 & 2^2 & 3^2 & 4^2 \\ 2^2 & 3^2 & 4^2 & 5^2 \\ 3^2 & 4^2 & 5^2 & 6^2 \\ 4^2 & 5^2 & 6^2 & 7^2 \end{bmatrix} = \begin{bmatrix} 1 & 4 & 9 & 16 \\ 4 & 9 & 16 & 25 \\ 9 & 16 & 25 & 36 \\ 16 & 25 & 36 & 49 \end{bmatrix}$$

Solution (Simplification): Apply $R_2 \rightarrow R_2 - R_1$, $R_3 \rightarrow R_3 - R_2$, $R_4 \rightarrow R_4 - R_3$:

$$A \sim \begin{bmatrix} 1 & 4 & 9 & 16 \\ 3 & 5 & 7 & 9 \\ 5 & 7 & 9 & 11 \\ 7 & 9 & 11 & 13 \end{bmatrix}$$

Apply $R_3 \rightarrow R_3 - R_2$, $R_4 \rightarrow R_4 - R_3$:

$$A \sim \begin{bmatrix} 1 & 4 & 9 & 16 \\ 3 & 5 & 7 & 9 \\ 2 & 2 & 2 & 2 \\ 2 & 2 & 2 & 2 \end{bmatrix}$$

Apply $R_4 \rightarrow R_4 - R_3$ and $R_3 \rightarrow R_3/2$:

$$A \sim \begin{bmatrix} 1 & 4 & 9 & 16 \\ 3 & 5 & 7 & 9 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Converting to Echelon Form: Swap R_1 and R_3 ($R_1 \leftrightarrow R_3$):

$$A \sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 3 & 5 & 7 & 9 \\ 1 & 4 & 9 & 16 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Apply $R_2 \rightarrow R_2 - 3R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$A \sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 4 & 6 \\ 0 & 3 & 8 & 15 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Divide R_2 by 2 ($R_2 \rightarrow R_2/2$):

$$A \sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 3 & 8 & 15 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Apply $R_3 \rightarrow R_3 - 3R_2$:

$$A \sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 2 & 6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Divide R_3 by 2 ($R_3 \rightarrow R_3/2$):

$$A \sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

- **Number of non-zero rows:** 3
- **Rank:** $\rho(A) = 3$
- **Nullity:** Order - Rank = $4 - 3 = 1$

3.4 3. Problem 3: Rank of a Rectangular Matrix

Problem: Find the rank and nullity of the 3×4 matrix:

$$A = \begin{bmatrix} 1 & 3 & 4 & 3 \\ 3 & 9 & 12 & 9 \\ 1 & 3 & 4 & 1 \end{bmatrix}$$

Solution: Apply $R_2 \rightarrow R_2 - 3R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$A \sim \begin{bmatrix} 1 & 3 & 4 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2 \end{bmatrix}$$

Divide R_3 by -2 ($R_3 \rightarrow R_3 / -2$) and swap R_2 and R_3 ($R_2 \leftrightarrow R_3$):

$$A \sim \begin{bmatrix} 1 & 3 & 4 & 3 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

- **Number of non-zero rows:** 2
- **Rank:** $\rho(A) = 2$
- (Note: Nullity is strictly defined for square matrices, but conceptually using the maximum square submatrix order, it is $3 - 2 = 1$.)

3.5 4. Problem 4: Rank of another 4×4 Matrix

Problem: Find the rank and nullity of:

$$A = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 1 & 3 & 2 & 2 \\ 2 & 4 & 3 & 4 \\ 3 & 7 & 4 & 6 \end{bmatrix}$$

Solution: Apply $R_2 \rightarrow R_2 - R_1$, $R_3 \rightarrow R_3 - 2R_1$, $R_4 \rightarrow R_4 - 3R_1$:

$$A \sim \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

Apply $R_4 \rightarrow R_4 - R_2$:

$$A \sim \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

- **Number of non-zero rows:** 3
- **Rank:** $\rho(A) = 3$
- **Nullity:** Order $-$ Rank $= 4 - 3 = 1$

4 Lecture 04: Rank of Matrix By Normal / Canonical Form

Series: Matrix Theory **Lecture:** 04 of 09 **Video:** <https://www.youtube.com/watch?v=gXwEtwX8g8M>

4.1 Prerequisites

- Knowledge of matrix rank.
- Understanding of elementary row and column operations.

4.2 1. Normal (Canonical) Form Review

Every non-zero matrix A of rank r can be reduced, by a sequence of elementary row **and** column operations, to one of the following forms:

$$\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} I_r \\ 0 \end{bmatrix}, \quad [I_r \ 0], \quad \text{or} \quad I_r$$

where I_r represents an identity matrix of order r , and 0 represents a zero matrix. The value r gives the rank of the original matrix A .

4.3 2. Problem 1: Identical Rows

Problem: Find the rank and nullity of:

$$A = \begin{bmatrix} 1 & a & b & 0 \\ 0 & c & d & 1 \\ 1 & a & b & 0 \\ 0 & c & d & 1 \end{bmatrix}$$

Solution: We can easily spot identical rows. Apply row operations $R_3 \rightarrow R_3 - R_1$ and $R_4 \rightarrow R_4 - R_2$:

$$A \sim \begin{bmatrix} 1 & a & b & 0 \\ 0 & c & d & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The matrix is now in Echelon form with 2 non-zero rows. * **Rank:** $\rho(A) = 2$ * **Nullity:** Order $-$ Rank $= 4 - 2 = 2$

4.4 3. Problem 2: Normal Form of a 4×4 Matrix

Problem: Find the rank of the matrix using Normal Form:

$$A = \begin{bmatrix} -2 & -1 & -3 & -1 \\ 1 & 2 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$$

Solution: Step-by-step Row Operations: Swap $R_1 \leftrightarrow R_3$:

$$A \sim \begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 2 & -3 & -1 \\ -2 & -1 & -3 & -1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$$

Apply $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 + 2R_1$:

$$A \sim \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 2 & -4 & -2 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$$

Divide R_2 by 2 ($R_2 \rightarrow R_2/2$):

$$A \sim \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -2 & -1 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$$

Apply $R_3 \rightarrow R_3 + R_2$ and $R_4 \rightarrow R_4 - R_2$:

$$A \sim \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -2 & -1 \\ 0 & 0 & -3 & 0 \\ 0 & 0 & 3 & 0 \end{bmatrix}$$

Apply $R_4 \rightarrow R_4 + R_3$ and divide R_3 by -3 ($R_3 \rightarrow R_3/-3$):

$$A \sim \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -2 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

(The matrix is now in Echelon form, and we can already see the rank is 3. We now proceed to Normal Form using column operations.)

Step-by-step Column Operations: Apply $C_3 \rightarrow C_3 - C_1$ and $C_4 \rightarrow C_4 - C_1$:

$$A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -2 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Apply $C_3 \rightarrow C_3 + 2C_2$ and $C_4 \rightarrow C_4 + C_2$:

$$A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

This is the Normal Form block matrix:

$$A \sim \begin{bmatrix} I_3 & 0 \\ 0 & 0 \end{bmatrix}$$

- **Rank:** $\rho(A) = 3$
- **Nullity:** Order $-$ Rank $= 4 - 3 = 1$

4.5 4. Problem 3: Normal Form of another 4×4 Matrix

Problem: Find the rank and nullity of:

$$A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

Solution: Step-by-step Row Operations: Swap $R_1 \leftrightarrow R_2$:

$$A \sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 2 & 3 & -1 & -1 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

Apply $R_2 \rightarrow R_2 - 2R_1$, $R_3 \rightarrow R_3 - 3R_1$, $R_4 \rightarrow R_4 - 6R_1$:

$$A \sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 5 & 3 & 7 \\ 0 & 4 & 9 & 10 \\ 0 & 9 & 12 & 17 \end{bmatrix}$$

Apply $R_2 \rightarrow R_2 - R_3$:

$$A \sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 1 & -6 & -3 \\ 0 & 4 & 9 & 10 \\ 0 & 9 & 12 & 17 \end{bmatrix}$$

Apply $R_3 \rightarrow R_3 - 4R_2$ and $R_4 \rightarrow R_4 - 9R_2$:

$$A \sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 1 & -6 & -3 \\ 0 & 0 & 33 & 22 \\ 0 & 0 & 66 & 44 \end{bmatrix}$$

Apply $R_4 \rightarrow R_4 - 2R_3$ and divide R_3 by 11 ($R_3 \rightarrow R_3/11$):

$$A \sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 1 & -6 & -3 \\ 0 & 0 & 3 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Step-by-step Column Operations: Apply $C_2 \rightarrow C_2 + C_1$, $C_3 \rightarrow C_3 + 2C_1$, $C_4 \rightarrow C_4 + 4C_1$:

$$A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -6 & -3 \\ 0 & 0 & 3 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Apply $C_3 \rightarrow C_3 + 6C_2$ and $C_4 \rightarrow C_4 + 3C_2$:

$$A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 3 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Divide C_3 by 3 ($C_3 \rightarrow C_3/3$):

$$A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Apply $C_4 \rightarrow C_4 - 2C_3$:

$$A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

This gives the Normal Form:

$$A \sim \begin{bmatrix} I_3 & 0 \\ 0 & 0 \end{bmatrix}$$

- **Rank:** $\rho(A) = 3$
- **Nullity:** $\text{Order} - \text{Rank} = 4 - 3 = 1$

5 Lecture 05: Consistency of System of Linear Equations

Series: Matrix Theory **Lecture:** 05 of 09 **Video:** <https://www.youtube.com/watch?v=bAJLtUzhBQI>

5.1 Prerequisites

- Proficiency in calculating the rank of a matrix.
- Understanding of Echelon form reductions.

5.2 1. Recalling School Concepts: 2 Simultaneous Equations

Before generalizing to matrices, let's recall the algebraic conditions for a system of 2 simultaneous linear equations in two variables:

$$a_1x + b_1y + c_1 = 0$$

$$a_2x + b_2y + c_2 = 0$$

Three Algebraic Conditions: * **Case 1: Unique Solution** (Lines intersect)

\$\$

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$$

- **Case 2: No Solution** (Lines are parallel)

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

- **Case 3: Infinite Solutions** (Lines are coincident)

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

5.2.1 Determinant Link for Case 1

The unique solution condition directly links to matrices. If we rearrange:

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2} \implies a_1b_2 - a_2b_1 \neq 0$$

This is the determinant of the coefficient matrix:

$$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \neq 0$$

5.3 2. Terminology and System Representation

5.3.1 Definitions

- **Consistent System:** A system of equations having one or more valid solutions (i.e., it has either a unique solution or infinite solutions).
- **Inconsistent System:** A system of equations having absolutely no solution.

5.3.2 System of 3 Equations in 3 Variables

Consider the standard system:

$$a_1x + b_1y + c_1z = d_1$$

$$a_2x + b_2y + c_2z = d_2$$

$$a_3x + b_3y + c_3z = d_3$$

This system can be written compactly in matrix form as $AX = B$:

$$\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

- **Homogeneous vs. Non-homogeneous:**

- **Non-homogeneous system:** When the constant matrix B is not entirely zero ($B \neq O$). The system is $AX = B$.
- **Homogeneous system:** When the constant matrix B is the zero matrix ($B = O$). The system is $AX = O$.

5.3.3 The Augmented Matrix

The augmented matrix is formed by combining the coefficient matrix A and the constant column matrix B , separated by a line:

$$[A : B] = \left[\begin{array}{ccc|c} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \end{array} \right]$$

The order of this augmented matrix is 3×4 .

5.4 3. Working Rule to Test Consistency ($AX = B$)

To test the consistency of a non-homogeneous system of linear equations using matrix rank, follow this procedure:

1. Write the given system of equations in matrix form $AX = B$.
2. Form the augmented matrix $[A : B]$.
3. Transform the augmented matrix $[A : B]$ into **Echelon form** using row operations only.
4. Find the rank of the coefficient matrix A , denoted as $\rho(A)$, and the rank of the augmented matrix $[A : B]$, denoted as $\rho([A : B])$.
5. Apply the following criteria to conclude:
 - **Case 3.1:** If $\rho(A) \neq \rho([A : B])$, the system is **Inconsistent** (No Solution).
 - **Case 3.2:** If $\rho(A) = \rho([A : B])$, the system is **Consistent**.
 - **Subcase 3.2 (a):** If $\rho(A) = \rho([A : B]) = n$ (where n is the total number of unknown variables), the system has a **Unique Solution**.
 - **Subcase 3.2 (b):** If $\rho(A) = \rho([A : B]) = r < n$, the system has **Infinite Solutions**. To solve it, we must choose $n - r$ variables arbitrarily.

5.5 What Comes Next

6 Lecture 06: Questions Consistency of System of Linear Equations

Series: Matrix Theory Lecture: 06 of 09 Video: <https://www.youtube.com/watch?v=sV3-43kkvSo>

6.1 Prerequisites

- Understanding of the conditions for consistency (Rouché-Capelli theorem).
- Proficiency in reducing augmented matrices to Echelon form.

6.2 1. Problem 1: Unique Solution

Problem: Test the consistency of the system of equations and solve if consistent:

$$3x + y + 2z = 11$$

$$2x + 3y + z = 11$$

$$x + 2y + 3z = 14$$

Solution: Augmented Matrix representation:

$$[A : B] = \left[\begin{array}{ccc|c} 3 & 1 & 2 & 11 \\ 2 & 3 & 1 & 11 \\ 1 & 2 & 3 & 14 \end{array} \right]$$

Swap $R_1 \leftrightarrow R_3$ to get a leading 1:

$$[A : B] \sim \left[\begin{array}{ccc|c} 1 & 2 & 3 & 14 \\ 2 & 3 & 1 & 11 \\ 3 & 1 & 2 & 11 \end{array} \right]$$

Apply $R_2 \rightarrow R_2 - 2R_1$ and $R_3 \rightarrow R_3 - 3R_1$:

$$[A : B] \sim \left[\begin{array}{ccc|c} 1 & 2 & 3 & 14 \\ 0 & -1 & -5 & -17 \\ 0 & -5 & -7 & -31 \end{array} \right]$$

Apply $R_2 \rightarrow -R_2$:

$$[A : B] \sim \left[\begin{array}{ccc|c} 1 & 2 & 3 & 14 \\ 0 & 1 & 5 & 17 \\ 0 & -5 & -7 & -31 \end{array} \right]$$

Apply $R_3 \rightarrow R_3 + 5R_2$:

$$[A : B] \sim \left[\begin{array}{ccc|c} 1 & 2 & 3 & 14 \\ 0 & 1 & 5 & 17 \\ 0 & 0 & 18 & 54 \end{array} \right]$$

Apply $R_3 \rightarrow R_3/18$:

$$[A : B] \sim \left[\begin{array}{ccc|c} 1 & 2 & 3 & 14 \\ 0 & 1 & 5 & 17 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

Ranks and Solution: * $\rho(A) = 3$ and $\rho([A : B]) = 3$. * Since $\rho(A) = \rho([A : B]) = 3$ (number of unknowns), the system is consistent and has a unique solution.

Back Substitution:

$$z = 3$$

$$y + 5(3) = 17 \implies y = 2$$

$$x + 2(2) + 3(3) = 14 \implies x = 1$$

Unique Solution: $x = 1, y = 2, z = 3$.

6.3 2. Problem 2: 4 Equations, 3 Unknowns

Problem: Test the consistency of the system of equations and solve if consistent:

$$x + 2y - z = 3$$

$$3x - y + 2z = 1$$

$$2x - 2y + 3z = 2$$

$$x - y + z = -1$$

Solution: Augmented Matrix representation:

$$[A : B] = \left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 3 & -1 & 2 & 1 \\ 2 & -2 & 3 & 2 \\ 1 & -1 & 1 & -1 \end{array} \right]$$

Apply $R_2 \rightarrow R_2 - 3R_1$, $R_3 \rightarrow R_3 - 2R_1$, and $R_4 \rightarrow R_4 - R_1$:

$$[A : B] \sim \left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & -7 & 5 & -8 \\ 0 & -6 & 5 & -4 \\ 0 & -3 & 2 & -4 \end{array} \right]$$

(Note: The lecturer warns against arithmetic mistakes here which can lead to falsely concluding the system is inconsistent. We follow the correct, robust path).

Apply $R_2 \rightarrow -(R_2 - 2R_4)$ and $R_3 \rightarrow R_3 - 2R_4$:

$$[A : B] \sim \left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 4 \\ 0 & -3 & 2 & -4 \end{array} \right]$$

Apply $R_4 \rightarrow R_4 + 3R_2$:

$$[A : B] \sim \left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & -1 & -4 \end{array} \right]$$

Apply $R_4 \rightarrow R_4 + R_3$:

$$[A : B] \sim \left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Ranks and Solution: * Rank of A is $\rho(A) = 3$, rank of $[A : B] = 3$. * Consistent system with a unique solution.

Back Substitution:

$$z = 4$$

$$y - 4 = 0 \implies y = 4$$

$$x + 2(4) - 4 = 3 \implies x = -1$$

Unique Solution: $x = -1, y = 4, z = 4$.

6.4 3. Problem 3: Infinitely Many Solutions

Problem: Test the consistency and solve:

$$5x + 3y + 7z = 4$$

$$3x + 26y + 2z = 9$$

$$7x + 2y + 10z = 5$$

Solution: Augmented Matrix representation:

$$[A : B] = \left[\begin{array}{ccc|c} 5 & 3 & 7 & 4 \\ 3 & 26 & 2 & 9 \\ 7 & 2 & 10 & 5 \end{array} \right]$$

Apply $R_1 \rightarrow 2R_2 - R_1$:

$$[A : B] \sim \left[\begin{array}{ccc|c} 1 & 49 & -3 & 14 \\ 3 & 26 & 2 & 9 \\ 7 & 2 & 10 & 5 \end{array} \right]$$

Apply $R_2 \rightarrow R_2 - 3R_1$ and $R_3 \rightarrow R_3 - 7R_1$:

$$[A : B] \sim \left[\begin{array}{ccc|c} 1 & 49 & -3 & 14 \\ 0 & -121 & 11 & -33 \\ 0 & -341 & 31 & -93 \end{array} \right]$$

Apply $R_2 \rightarrow -R_2/11$ and $R_3 \rightarrow -R_3/31$:

$$[A : B] \sim \left[\begin{array}{ccc|c} 1 & 49 & -3 & 14 \\ 0 & 11 & -1 & 3 \\ 0 & 11 & -1 & 3 \end{array} \right]$$

Apply $R_3 \rightarrow R_3 - R_2$:

$$[A : B] \sim \left[\begin{array}{ccc|c} 1 & 49 & -3 & 14 \\ 0 & 11 & -1 & 3 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Ranks and Solution: * $\rho(A) = \rho([A : B]) = 2 < 3$ (number of unknowns). * Consistent system with infinitely many solutions.

Parametric Formulation: From R_2 : $11y - z = 3$. Let $z = k$ (where $k \in \mathbb{R}$).

$$y = \frac{k+3}{11}$$

From R_1 : $x + 49y - 3z = 14$.

$$x = 14 - 49\left(\frac{k+3}{11}\right) + 3k \implies x = \frac{7-16k}{11}$$

Infinite Solutions: $x = \frac{7-16k}{11}, y = \frac{k+3}{11}, z = k$.

6.5 4. Problem 4: Parameter Analysis (λ, μ)

Problem: For what values of λ and μ do the equations have: (i) no solution, (ii) a unique solution, (iii) infinitely many solutions?

$$x + y + z = 6$$

$$x + 2y + 3z = 10$$

$$x + 2y + \lambda z = \mu$$

Solution: Augmented Matrix representation:

$$[A : B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \lambda & \mu \end{array} \right]$$

Apply $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$[A : B] \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & \lambda - 1 & \mu - 6 \end{array} \right]$$

Apply $R_3 \rightarrow R_3 - R_2$:

$$[A : B] \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & \lambda - 3 & \mu - 10 \end{array} \right]$$

Consistency Analysis: 1. **Case 1: No Solution** Requires $\rho(A) \neq \rho([A : B])$. This happens if $\rho(A) = 2$ and $\rho([A : B]) = 3$.

$$\lambda - 3 = 0 \implies \lambda = 3$$

$$\mu - 10 \neq 0 \implies \mu \neq 10$$

2. **Case 2: Unique Solution** Requires $\rho(A) = \rho([A : B]) = 3$.

$$\lambda - 3 \neq 0 \implies \lambda \neq 3$$

μ can be any real value ($\mu \in \mathbb{R}$).

3. **Case 3: Infinitely Many Solutions** Requires $\rho(A) = \rho([A : B]) = 2 < 3$.

$$\lambda - 3 = 0 \implies \lambda = 3$$

$$\mu - 10 = 0 \implies \mu = 10$$

7 Lecture 07: Concept of Eigen Values and Vectors

Series: Matrix Theory Lecture: 07 of 09 Video: <https://www.youtube.com/watch?v=GyT-SdFtqq4>

7.1 Prerequisites

- Familiarity with matrix determinants and polynomials.
- Ability to solve homogeneous systems of linear equations.

7.2 1. Definitions

7.2.1 Characteristic Matrix

Let A be a square matrix of order n , let λ be a scalar (unknown variable), and let I be the identity matrix of the same order as A . The matrix:

$$A - \lambda I$$

is called the characteristic matrix of A .

Example: If $A = \begin{bmatrix} 2 & 3 \\ 4 & 6 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then:

$$A - \lambda I = \begin{bmatrix} 2 & 3 \\ 4 & 6 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 - \lambda & 3 \\ 4 & 6 - \lambda \end{bmatrix}$$

7.2.2 Characteristic Polynomial & Equation

- **Characteristic Polynomial:** The determinant of the characteristic matrix, $|A - \lambda I|$, which yields a polynomial in λ .
- **Characteristic Equation:** Equating the characteristic polynomial to zero:

$$|A - \lambda I| = 0$$

7.3 2. Eigenvalues and Eigenvectors

7.3.1 Characteristic Roots / Eigenvalues

The roots of the characteristic equation $|A - \lambda I| = 0$ are called characteristic roots, **eigenvalues**, or latent values of matrix A . These values are unique for a given matrix.

7.3.2 Eigenvectors

Let A be a square matrix of order n , and λ be an eigenvalue of A . A **non-zero** column vector:

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

satisfying the relation:

$$(A - \lambda I)X = 0$$

is called an **eigenvector** of A corresponding to the eigenvalue λ .

Key Properties: 1. $(A - \lambda I)X = 0$ is a homogeneous system of linear equations. 2. The eigenvector X must be non-zero ($X \neq 0$). 3. The eigenvector corresponding to an eigenvalue is **not unique**. For any scalar $k \neq 0$, kX is also an eigenvector.

7.4 3. Working Rule

1. **Step 1:** Find the characteristic equation $|A - \lambda I| = 0$.
2. **Step 2:** Solve the characteristic equation to find eigenvalues $\lambda = \lambda_1, \lambda_2, \dots, \lambda_n$.
3. **Step 3:** For each eigenvalue λ_i , solve the homogeneous system $(A - \lambda_i I)X = 0$ using row operations to reduce the matrix to Echelon form, and choose arbitrary variables to find a non-zero eigenvector X .

7.5 4. Solved Problem 1

Problem: Find the eigenvalues and eigenvectors of the matrix:

$$A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$$

Step 1: Characteristic Equation

$$|A - \lambda I| = 0 \implies \begin{vmatrix} 5 - \lambda & 4 \\ 1 & 2 - \lambda \end{vmatrix} = 0$$

Step 2: Solve the determinant

$$\begin{aligned} (5 - \lambda)(2 - \lambda) - 4 &= 0 \\ 10 - 5\lambda - 2\lambda + \lambda^2 - 4 &= 0 \\ \lambda^2 - 7\lambda + 6 &= 0 \end{aligned}$$

Factoring:

$$\begin{aligned} \lambda^2 - 6\lambda - \lambda + 6 &= 0 \\ \lambda(\lambda - 6) - 1(\lambda - 6) &= 0 \\ (\lambda - 6)(\lambda - 1) &= 0 \end{aligned}$$

Eigenvalues: $\lambda = 1, 6$.

Step 3: Eigenvectors

Case 1: Eigenvector for $\lambda = 1$ Let the eigenvector be $X_1 = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$. Solve $(A - 1I)X_1 = 0$:

$$\begin{bmatrix} 5 - 1 & 4 \\ 1 & 2 - 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies \begin{bmatrix} 4 & 4 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Apply row operation $R_1 \rightarrow \frac{1}{4}R_1$:

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Apply row operation $R_2 \rightarrow R_2 - R_1$:

$$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This gives the equation $x_1 + x_2 = 0 \implies x_2 = -x_1$. Choosing $x_1 = 1$ arbitrarily ($x_1 \neq 0$), we get $x_2 = -1$.

$$X_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

(General form is $X_1 = k \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ for $k \neq 0$).

Case 2: Eigenvector for $\lambda = 6$ Let the eigenvector be $X_2 = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$. Solve $(A - 6I)X_2 = 0$:

$$\begin{bmatrix} 5 - 6 & 4 \\ 1 & 2 - 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies \begin{bmatrix} -1 & 4 \\ 1 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Apply row operation $R_2 \rightarrow R_2 + R_1$:

$$\begin{bmatrix} -1 & 4 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This gives the equation $-x_1 + 4x_2 = 0 \implies x_1 = 4x_2$. Choosing $x_2 = 1$ arbitrarily ($x_2 \neq 0$), we get $x_1 = 4$.

$$X_2 = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

(General form is $X_2 = k \begin{bmatrix} 4 \\ 1 \end{bmatrix}$ for $k \neq 0$).

8 Lecture 08: Cayley Hamilton Theorem: State and Prove

Series: Matrix Theory **Lecture:** 08 of 09 (Note: Listed as Lecture-9 in some course materials) **Video:** <https://www.youtube.com/watch?v=yJUoUn9f2Wo>

8.1 Prerequisites

- Knowledge of characteristic polynomials and characteristic equations.
- Familiarity with matrix cofactors and the adjoint matrix identity $D \cdot \text{adj}(D) = |D|I$.

8.2 1. Statement of the Theorem

Cayley-Hamilton Theorem: Every square matrix satisfies its own characteristic equation.

Mathematical Form: Let A be a square matrix of order $n \times n$. Let its characteristic equation be:

$$|A - \lambda I| = a_0 \lambda^n + a_1 \lambda^{n-1} + a_2 \lambda^{n-2} + \dots + a_n = 0$$

Then, replacing the scalar variable λ with the matrix A yields a valid matrix identity:

$$a_0 A^n + a_1 A^{n-1} + a_2 A^{n-2} + \dots + a_n I = O$$

where I is the identity matrix of order n , and O is the zero matrix of order n .

8.3 2. Formal Proof

8.3.1 Step 1: Defining the Adjoint Polynomial

Let A be an $n \times n$ square matrix with the characteristic equation:

$$|A - \lambda I| = a_0 \lambda^n + a_1 \lambda^{n-1} + \dots + a_n = 0$$

The elements of the characteristic matrix $(A - \lambda I)$ are polynomials in λ of degree at most 1. The elements of its adjoint matrix, $\text{adj}(A - \lambda I)$, are the cofactors of $(A - \lambda I)$. Since cofactors are determinants of order $n - 1$, they are polynomials in λ of degree at most $n - 1$.

Therefore, we can express $\text{adj}(A - \lambda I)$ as a matrix polynomial in λ :

$$\text{adj}(A - \lambda I) = B_0 \lambda^{n-1} + B_1 \lambda^{n-2} + B_2 \lambda^{n-3} + \dots + B_{n-1}$$

where $B_0, B_1, \dots, B_{n-1}$ are $n \times n$ matrices whose elements are constant scalars independent of λ .

8.3.2 Step 2: Applying the Adjoint Identity

We use the fundamental matrix property for any square matrix D :

$$D \cdot \text{adj}(D) = |D|I$$

Substituting $D = A - \lambda I$, we get:

$$(A - \lambda I) \cdot \text{adj}(A - \lambda I) = |A - \lambda I|I$$

Now, substitute the polynomial expressions into both sides:

$$(A - \lambda I)(B_0 \lambda^{n-1} + B_1 \lambda^{n-2} + \dots + B_{n-1}) = (a_0 \lambda^n + a_1 \lambda^{n-1} + \dots + a_n)I$$

8.3.3 Step 3: Equating Coefficients

Expand the left-hand side and equate the matrix coefficients of corresponding powers of λ on both sides:

- For λ^n :

$$-B_0 = a_0 I$$

- For λ^{n-1} :

$$AB_0 - B_1 = a_1 I$$

- For λ^{n-2} :

$$AB_1 - B_2 = a_2 I$$

- ... (continuing the pattern) ...

- For λ^1 :

$$AB_{n-2} - B_{n-1} = a_{n-1} I$$

- For the constant term (λ^0):

$$AB_{n-1} = a_n I$$

8.3.4 Step 4: Elimination and Conclusion

To eliminate the B_i matrices, pre-multiply these equations successively by $A^n, A^{n-1}, A^{n-2}, \dots, A, I$:

$$-A^n B_0 = a_0 A^n$$

$$A^n B_0 - A^{n-1} B_1 = a_1 A^{n-1}$$

$$A^{n-1} B_1 - A^{n-2} B_2 = a_2 A^{n-2}$$

...

$$A^2 B_{n-2} - AB_{n-1} = a_{n-1} A$$

$$AB_{n-1} = a_n I$$

Now, add all these equations together. Notice that on the left-hand side (LHS), the terms cancel each other out in pairs (telescoping sum):

$$-A^n B_0 + A^n B_0 - A^{n-1} B_1 + A^{n-1} B_1 - A^{n-2} B_2 + \dots - AB_{n-1} + AB_{n-1} = 0$$

The sum of the right-hand side (RHS) becomes:

$$a_0 A^n + a_1 A^{n-1} + a_2 A^{n-2} + \dots + a_{n-1} A + a_n I$$

Therefore, equating LHS and RHS gives:

$$a_0 A^n + a_1 A^{n-1} + a_2 A^{n-2} + \dots + a_n I = 0$$

Hence Proved.

8.4 What Comes Next

9 Lecture 09: Cayley Hamilton Theorem: Solved Problems

Series: Matrix Theory **Lecture:** 09 of 09 (Note: Listed as Lecture-10 in some course materials) **Video:** <https://www.youtube.com/watch?v=FwpmB9cGSLE>

9.1 Prerequisites

- Understanding of the Cayley-Hamilton theorem statement.
- Proficiency in matrix multiplication and characteristic equation derivation.

9.2 1. Application: Finding the Inverse of a Matrix

The most prominent application of the Cayley-Hamilton theorem is computing the inverse of a large matrix (A^{-1}) efficiently by bypassing the need to compute the cofactor and adjoint matrices.

According to the theorem, every square matrix satisfies its characteristic equation:

$$a_0 A^n + a_1 A^{n-1} + a_2 A^{n-2} + \dots + a_{n-1} A + a_n I = O$$

Assuming the matrix is non-singular ($|A| = a_n \neq 0$), we multiply both sides by A^{-1} :

$$a_0 A^{n-1} + a_1 A^{n-2} + a_2 A^{n-3} + \dots + a_{n-1} I + a_n A^{-1} = O$$

Rearranging the terms to isolate A^{-1} :

$$A^{-1} = -\frac{1}{a_n}(a_0 A^{n-1} + a_1 A^{n-2} + \dots + a_{n-1} I)$$

9.3 2. Problem 1: Order 2 Matrix

Problem: Verify the Cayley-Hamilton Theorem and find A^{-1} for:

$$A = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$$

1. Characteristic Equation:

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 2 \\ -1 & 3-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)(3-\lambda) + 2 = 0 \implies \lambda^2 - 4\lambda + 5 = 0$$

2. Verification: We must show $A^2 - 4A + 5I = O$. First, calculate A^2 :

$$A^2 = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} -1 & 8 \\ -4 & 7 \end{bmatrix}$$

Substitute into the equation:

$$A^2 - 4A + 5I = \begin{bmatrix} -1 & 8 \\ -4 & 7 \end{bmatrix} - \begin{bmatrix} 4 & 8 \\ -4 & 12 \end{bmatrix} + \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O$$

(Verified)

3. Finding A^{-1} : From $A^2 - 4A + 5I = O$, multiply by A^{-1} :

$$A - 4I + 5A^{-1} = O \implies A^{-1} = \frac{1}{5}(4I - A)$$

Calculate the inner term:

$$4I - A = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} - \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ 1 & 1 \end{bmatrix}$$

Therefore:

$$A^{-1} = \frac{1}{5} \begin{bmatrix} 3 & -2 \\ 1 & 1 \end{bmatrix}$$

9.4 3. Problem 2: Order 3 Matrix

Problem: Verify the Cayley-Hamilton Theorem and find A^{-1} for:

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$$

1. Characteristic Equation:

$$|A - \lambda I| = \begin{vmatrix} 1 - \lambda & 0 & 2 \\ 0 & 2 - \lambda & 1 \\ 2 & 0 & 3 - \lambda \end{vmatrix} = 0$$

Expanding along the first row:

$$(1 - \lambda)[(2 - \lambda)(3 - \lambda) - 0] + 2[0 - 2(2 - \lambda)] = 0$$

$$(1 - \lambda)(\lambda^2 - 5\lambda + 6) - 4(2 - \lambda) = 0$$

$$(6 - 5\lambda + \lambda^2 - 6\lambda + 5\lambda^2 - \lambda^3) - 8 + 4\lambda = 0$$

$$-\lambda^3 + 6\lambda^2 - 7\lambda - 2 = 0 \implies \lambda^3 - 6\lambda^2 + 7\lambda + 2 = 0$$

2. Verification: We must show $A^3 - 6A^2 + 7A + 2I = O$. First, calculate A^2 and A^3 :

$$A^2 = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 5 & 0 & 8 \\ 2 & 4 & 5 \\ 8 & 0 & 13 \end{bmatrix}$$

$$A^3 = A^2 \cdot A = \begin{bmatrix} 5 & 0 & 8 \\ 2 & 4 & 5 \\ 8 & 0 & 13 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 21 & 0 & 34 \\ 12 & 8 & 23 \\ 34 & 0 & 55 \end{bmatrix}$$

Substitute into the equation:

$$\begin{aligned} A^3 - 6A^2 + 7A + 2I &= \begin{bmatrix} 21 & 0 & 34 \\ 12 & 8 & 23 \\ 34 & 0 & 55 \end{bmatrix} - 6 \begin{bmatrix} 5 & 0 & 8 \\ 2 & 4 & 5 \\ 8 & 0 & 13 \end{bmatrix} + 7 \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} + 2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 21 - 30 + 7 + 2 & 0 & 34 - 48 + 14 \\ 12 - 12 & 8 - 24 + 14 + 2 & 23 - 30 + 7 \\ 34 - 48 + 14 & 0 & 55 - 78 + 21 + 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O \end{aligned}$$

(Verified)

3. Finding A^{-1} : From $A^3 - 6A^2 + 7A + 2I = O$, multiply by A^{-1} :

$$A^2 - 6A + 7I + 2A^{-1} = O \implies 2A^{-1} = -A^2 + 6A - 7I$$

$$A^{-1} = \frac{1}{2}(-A^2 + 6A - 7I)$$

Calculate the inner term:

$$-A^2 + 6A - 7I = - \begin{bmatrix} 5 & 0 & 8 \\ 2 & 4 & 5 \\ 8 & 0 & 13 \end{bmatrix} + 6 \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} - 7 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -6 & 0 & 4 \\ -2 & 1 & 1 \\ 4 & 0 & -2 \end{bmatrix}$$

Therefore:

$$A^{-1} = \frac{1}{2} \begin{bmatrix} -6 & 0 & 4 \\ -2 & 1 & 1 \\ 4 & 0 & -2 \end{bmatrix} = \begin{bmatrix} -3 & 0 & 2 \\ -1 & \frac{1}{2} & \frac{1}{2} \\ 2 & 0 & -1 \end{bmatrix}$$

9.5 What Comes Next

This concludes the Matrix Theory lecture series. Mastering these concepts provides the foundational tools necessary for advanced applications in linear algebra.